

1. First Palindromic String in an Array

Aim: To find the first palindromic string in an array and analyze its time complexity.

Algorithm

1. Read an array of strings.
2. Check each string one by one.
3. Compare the string with its reverse.
4. If both are equal, it is a palindrome.
5. Print the first palindrome found.
6. If no palindrome exists, print an empty string.

Python Program

```
words = input("Enter strings separated by space: ").split()
```

```
found = False
```

```
for word in words:
```

```
    if word == word[::-1]:
```

```
        print("First Palindromic String:", word)
```

```
        found = True
```

```
        break
```

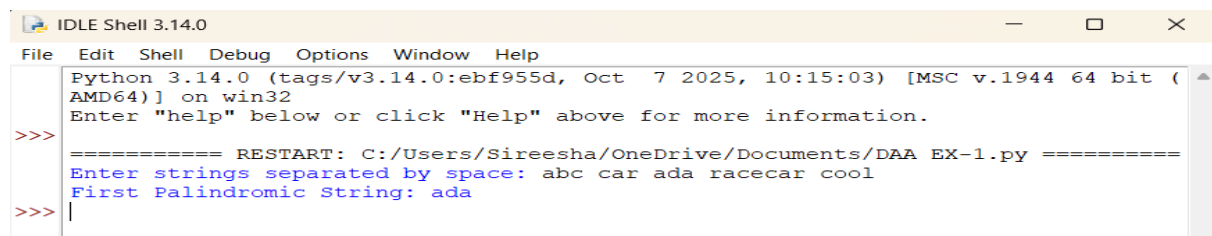
```
if not found:
```

```
    print("First Palindromic String: \"\")
```

Input

abc car ada racecar cool

Output



```
IDLE Shell 3.14.0
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/Sireesha/OneDrive/Documents/DAA EX-1.py =====
Enter strings separated by space: abc car ada racecar cool
First Palindromic String: ada
>>> |
```

Result

Thus, the Python program to find the first palindromic string in an array was successfully executed, and its time and space complexities were analyzed.